Electronics and Electrical Communications Engineering Department

Faculty of Engineering

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Neural Networks

Assignment 1
Classical Machine Learning Methods

4th Year
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1 Introduction

This report presents a semi-supervised pipeline for labeling a dataset of 10,000 handwritten digit images. The objective is to achieve high classification accuracy while minimizing manual labeling effort by combining clustering, feature extraction, and active learning.

2 Pipeline Overview

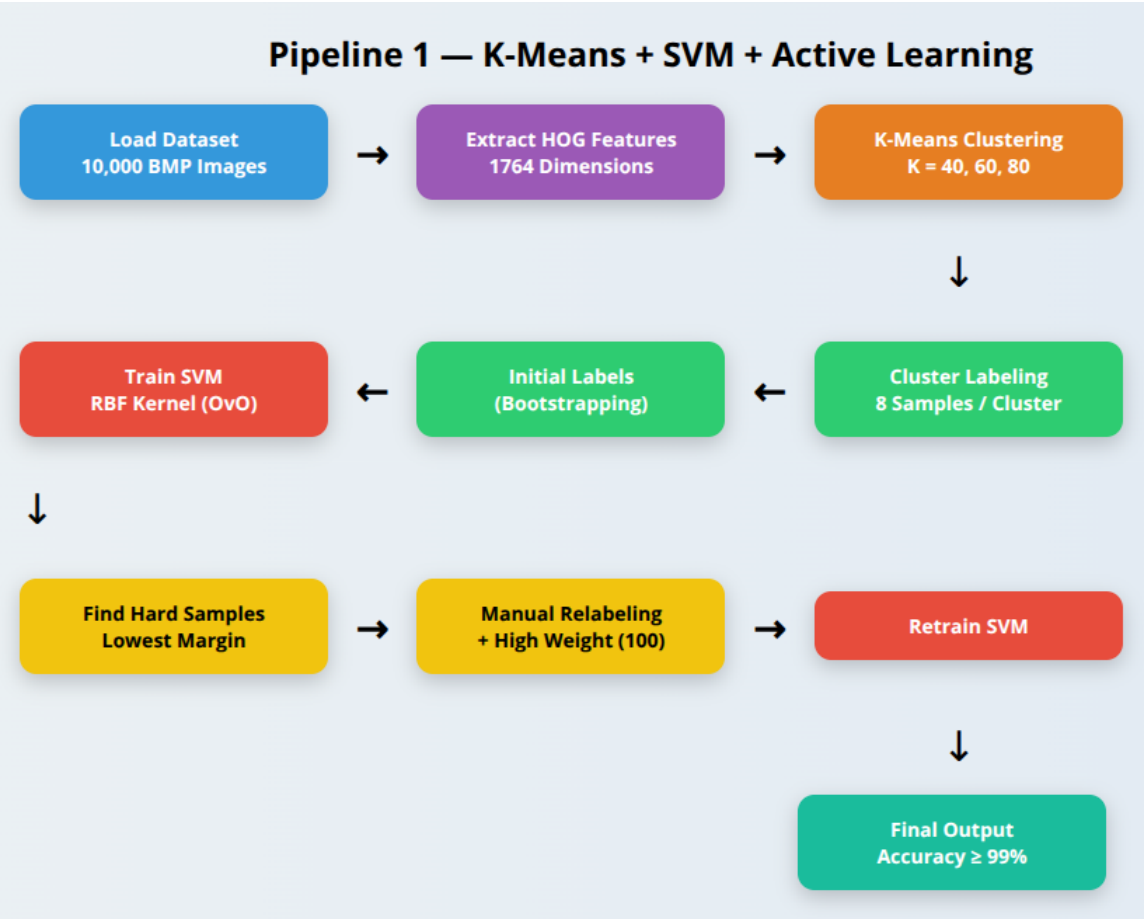

Figure 1: Pipeline 1: K-Means + SVM + Active Learning

The pipeline consists of the following steps:

- Load dataset (10,000 BMP images)
- Extract HOG features
- Apply K-Means clustering
- Label clusters (bootstrapping)
- Train SVM classifier
- Apply active learning for refinement

3 Feature Extraction

Histogram of Oriented Gradients (HOG) was used to extract features from images.

- Image size: 32×32
- Feature vector size: 1764

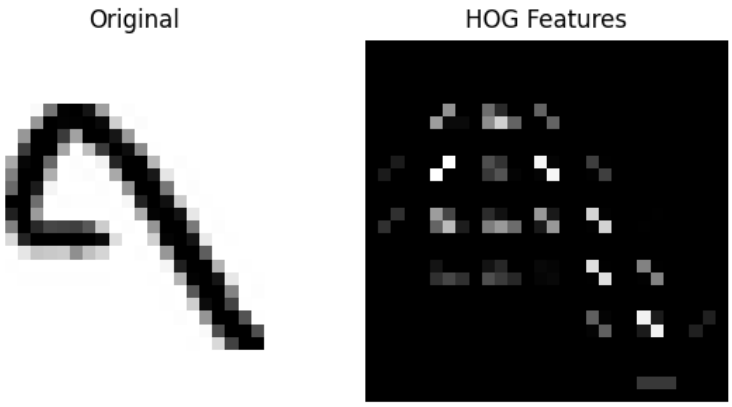

Figure 2: Example of HOG Feature Representation

4 K-Means Clustering

K-Means clustering was applied with different values of K:

- K = 40
- K = 60
- K = 80

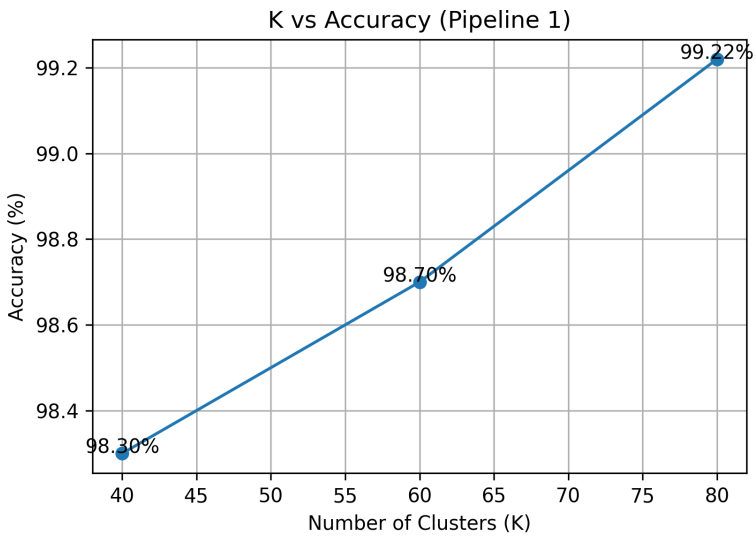

Figure 3: K vs Accuracy Before SVM

5 SVM and Active Learning

A multi-class SVM with RBF kernel (One-vs-One strategy) was used.
Active learning was applied as follows:

- Select 30 least confident samples (lowest margin)
- Relabel manually
- Assign weight = 100
- Retrain SVM

The process stops when:

- Accuracy $\geq 99\%$
- Improvement $< 0.1\%$

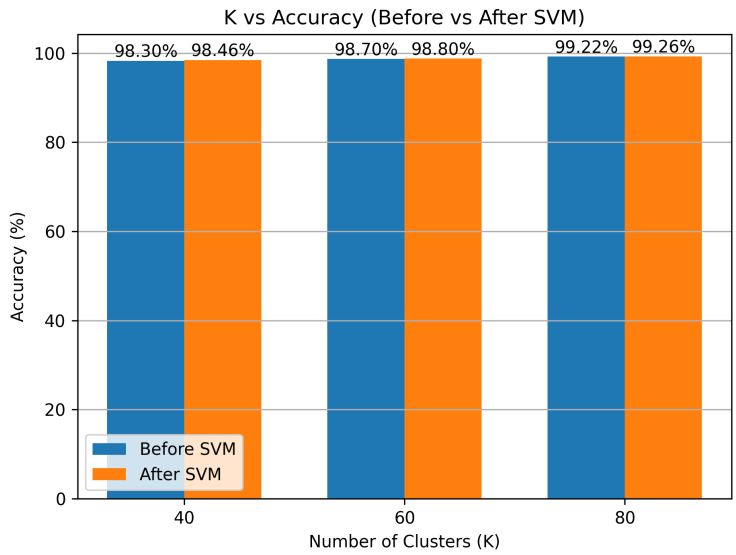

Figure 4: Accuracy Before vs After SVM

6 Results

6.1 Accuracy

K	Before SVM (%)	After SVM (%)
40	98.30	98.46
60	98.70	98.80
80	99.22	99.26

Table 1: Accuracy Comparison

6.2 Iterations

K	Iterations
40	2
60	2
80	2

Table 2: Number of Iterations

7 Manual Effort

For K = 80:

- Cluster labeling: $80 \times 8 = 640$ images
- Active learning: $30 \times 2 = 60$ images

Total images manually handled:

$$700 \text{ images}$$

Total time:

- Cluster labeling: 1600 seconds
- Active learning: 600 seconds

$$\text{Total time} = 2200 \text{ seconds} \approx 36.7 \text{ minutes}$$

8 Conclusion

The proposed pipeline significantly reduces manual labeling effort from 27.8 hours to approximately 37 minutes while achieving 99.26% accuracy.

K-Means clustering provides strong initial labels, while SVM and active learning refine ambiguous samples. The improvement from SVM is more noticeable at lower K values, while higher K values already produce highly pure clusters.